

KAREN ZHOU

Avon, CT | kzhou.wisc@gmail.com | (860) 877-6847 | US Citizen

SUMMARY

Biomedical Data Scientist pursuing a M.S. in Data Science in Biomedicine at UCLA (expected June 2026) and with hands-on experience in bioinformatics, AI for healthcare, and large-scale data analysis. Skilled in genome assembly, medical imaging algorithms, and electronic health record analysis. Proficient in Python, R, SQL, and machine learning, with a strong record of delivering scalable solutions in biotech and healthcare.

SKILLS

Programming: Python, R, Java, SQL, C, JavaScript, HTML

Data Science & Analytics: Pandas, NumPy, Matplotlib, Seaborn, NetworkX, QlikView, Neo4j

Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, Regression, Clustering, PCA, NLP, Random Forest, CNNs, GNNs

Bioinformatics & Biomedical: GWAS, PRS, Genome assembly, sequence alignment, medical imaging, ITK-SNAP.

Tools: Git, GitHub, Linux/command line, VSCode, PyCharm, PostgreSQL, BeautifulSoup

EDUCATION

University of California, Los Angeles – M.S., Data Science in Biomedicine (Jan 2025 – Present, Expected Graduation Date: June 2026)

Courses: Machine Learning Applications in Biomedicine, Medical Imaging in Data Science, Data Science Algorithms in Biomedicine, Large-Scale Social and Complex Networks, Large Scale Machine Learning, Digital Speech Processing, Recent Research in Data Science in Genomic Medicine

University of Wisconsin–Madison – B.S., Computer Science (Sep 2019 – Dec 2022)

Dean's List | Course: Bioinformatics, Software Engineering, Introduction to Algorithms, Discrete Mathematics, Operating Systems, Artificial Intelligence, Applied Statistics for Engineers, Programming, Computer Graphics, Machine Organization and Programming

RELEVANT EXPERIENCE

Pharmolix – Artificial Intelligence Intern (Jan 2024 – Dec 2024)

- Developed biomedical instruction datasets for large language model fine-tuning and helped with diverse and high quality instruction generation
- Applied and researched clustering, neural networks, and NLP to improve training efficiency for healthcare AI
- Contributed to a biotech R&D that is less than a year old

iCylon – Software Engineer Intern (Sept 2023 – Dec 2023)

- Developed Python scripts for web scraping (HTML/JavaScript parsing with BeautifulSoup) to automate data extraction. Designed ETL pipeline: stored raw data in PostgreSQL, transformed it to CSV, and loaded into Neo4j graph database for analysis.
- Conducted research and assisted in presentations to potential clients, explaining data-driven insights.
- Contributed to a startup

Pharmaron – Software Engineer Intern (Jun 2023 – Aug 2023)

- Automated Excel-to-Word data reporting for biotech workflows, significantly reducing processing time.
- Developed python scripts and excel templates to generate project data reports. Made sure the scripts can be run alone, without other classes. Used JSON, xlrd, and openpyxl to read and extract from data tables..

Navitus Health Solutions – *Data Analytics Intern* (Jun 2022 – Aug 2022)

- Used QlikView and SQL scripts to centralize healthcare claims data, increasing efficiency
- Assisted on reports for pharmacy benefit management analytics.

UCONN Health – *Lab Assistant Intern* (Jun 2018 – Aug 2018)

- Assisted in cancer immunotherapy experiments and data analysis (gel electrophoresis, antibody research).
- Rewarding experience in oncology lab

SELECTED PROJECTS

Genome Assembly Pipeline – *Algorithms in Biomedicine* (May 2025)

- Built genome assembler using de Bruijn graphs to reconstruct DNA sequences from short reads in fasta files
- Achieved accurate read position recovery using k-mer filtering and mapping; implemented full pipeline in Python.

Medical Imaging – *Medical Imaging in Data Science* (Mar 2025)

- Developed Python tools for medical imaging visualization and modality analysis.
- Created histogram-based image analysis and convolution-based filtering pipelines for MRI/CT data.
- Designed, trained, and evaluated CNNs for biomedical image classification and segmentation.
- Implemented image registration algorithms and conducted statistical testing on imaging datasets.

Pruning Scalable GNNs for Real-Time Fraud Detection – *Large-Scale Machine Learning* (December 2025)

- Co-authored research on scalable GNN pruning for fraud detection using SynFlow and biased neighborhood sampling
- Achieved 90–99% sparsity with near-baseline AUPRC and ~100× lower inference FLOPs on DGraph-Fin
- Implemented and evaluated sparse GraphSAGE models in PyTorch under extreme class imbalance

Asthma Exacerbation Prediction with EHR + Polygenic Risk Scores (Mar - Jun 2026)

- Built predictive models for asthma exacerbation risk using EHR-derived clinical features and GWAS-based polygenic risk scores (PRS)
- Engineered longitudinal diagnosis, biomarker, medication, and healthcare utilization features from UCLA clinical databases using SQL and Python
- Performed GWAS QC, PCA, and PRS generation using PLINK/PRSice and integrated genetic risk with clinical prediction models
- Evaluated predictive performance using ROC AUC, PR-AUC, calibration, and Brier score

LEADERSHIP

Captain/Member/MVP — Avon High School Varsity Swim Team (December 2015 - March 2019)

- Led team to multiple championships; record holder in 200 free relay. Represented the team at meets/events, and had additional responsibilities at meets, swim activities, and banquets.